ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)

ACADEX PRACTICE EXAMINATION

O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1

PAPER 1 November 2020

2 hours 30 minutes

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.
INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [ ].

1. Evaluate $3 + 4 \times 6 - 4$. **[2]**

2. Expand $(x + 1)(x + 1)$. **[3]**

3. Overnight in Gweru the temperature was -3°C. By 14:00 it was 32°C. Find the rise. **[2]**

4. In a class of Gifford High, $n(A) = 13$ (play soccer), $n(A \cap B) = 4$ (play both). How many play soccer only? **[2]**

5. Simplify 3^3 × 3^2. Leave as a power of 3. **[2]**

6. A map of Gweru uses scale 3 : 800000. A road is 12 cm on the map. Find the real length in km. (1 : 100 000 means 1 cm ↔ 1 km if 3:8 is first simplified — instead: scale 3 cm to 8 km.) **[3]**

7. A tank in Mkoba holds 42 litres. It is 2/6 full. How many litres are in the tank? **[3]**

8. Find 15% of 40. **[3]**

9. Find the HCF and LCM of 48 and 20. **[4]**

10. Write 90000 in standard form. **[3]**

11. Factorise $x^2 - 9$. **[4]**

12. Area of a circle radius 7 cm. Take $\pi = 22/7$. **[4]**

13. A trader at Mkoba buys a bag of potatoes for $80 and sells for $88. Percentage profit? **[4]**

14. A cylindrical drum has radius 7 cm and height 4 cm. Find its volume. $\pi = 22/7$. **[4]**

15. Find the gradient of the line joining (2, 3) and (4, 11). **[3]**

16. Factorise $x^2 + 5x + 6$. **[4]**

17. Find the midpoint of (2, 6) and (2, 2). **[3]**

18. A regular polygon has 9 sides. Find each interior angle. **[4]**

19. A sequence is 7, 11, 15, 19, … Find an expression for the nth term. **[4]**

20. Find the mean of 12, 8, 7, 6, 9. **[3]**

21. In right-angled triangle PQR, $\hat{P}$ = 30° and hypotenuse PR = 10 cm. Find PQ, opposite 30°. **[4]**

22. Make r the subject of C = 2πr. **[4]**

23. A bag contains 7 red, 6 blue and 5 green beads. P(not green)? **[3]**

24. Work out (4 1 ; 4 2)(5 ; 2). **[4]**

25. A right-angled triangle has shorter sides 10 cm and 24 cm. Find the hypotenuse. **[4]**

26. Solve 4x + 2 < 18. **[3]**

27. Solve 3(x + 2) = 18. **[4]**

28. AB ∥ CD. A transversal makes an acute angle of 66° with AB. Find the co-interior angle on CD. **[3]**

29. Solve<<
>>x + y = 7<<
>>x − y = 1. **[4]**

30. A triangular plot has base 6 m and perpendicular height 11 m. Find its area. **[3]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).